When Does Pooling Make Intersectional Fairness Audits Falsely Reassuring?

ICT 5143: AFGML Capstone | Problem 8: Reliable intersectional fairness audits

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https://github.com/sumedhnvda/ict5143-p8-fairness-audits

Research question and decision context

- **Context:** an agency can contact only 30% of a list with *hypothetical* information outreach about public coverage. A Bayesian network ranks people; the top 30% are contacted.
- **Stakeholders:** people without public coverage who are never contacted, often in small race×sex×disability groups; the auditor who must sign off on the rule.
- $Y = 1$: **no public coverage** in the ACS record (not "uninsured", not "eligible"). $D = 1$: selected for outreach.
- **Question:** when does hierarchical pooling make an audit say $\Delta\text{EOpp} = \max_g \text{TPR}_g - \min_g \text{TPR}_g \leq 0.10$ when the rule is unfair?
- **H1:** when λ is chosen on validation and a pooled audit certifies ≤ 0.10, the frozen rule exceeds 0.10 on test in **more than half** of cases. Refuted if at most half fail on Texas.

Closest work and proposed contribution

- **Established:** Bayesian hierarchical estimation of intersectional fairness (Foulds et al., SDM 2020); structured regression for tiny subgroups (Herlihy et al., FAccT 2024); ϵ-contamination priors (Berger & Berliner 1986); marginal fairness can hide subgroup unfairness (Kearns et al., ICML 2018).
- **Open issue:** whether pooled *audit criteria* produce false "gap $\leq \epsilon$" claims when used to choose a decision rule on a real sparse design. We found no paper that tests this.
- **Contribution:** (i) a controlled evaluation of when pooling masks sparse disparities (H1); (ii) a fix: **design-matched calibration** plus **held-out certification** (H2a–c). We claim no new prior and no new metric.
- **Evidence needed:** out-of-sample Texas test of H1; paired ΔJ against every established audit; an ablation without design matching and without held-out certification.

19-paper comparison table: `writeup/literature_review.pdf`

Data and evaluation population

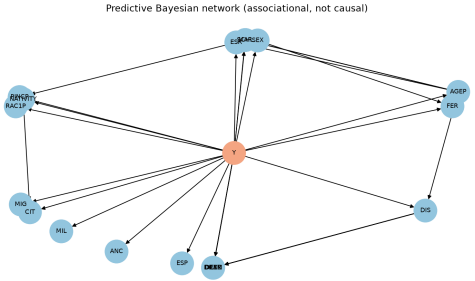
- Folktables ACSPublicCoverage, ACS PUMS 2018 1-year; CA core, TX replication.
- Filter AGEP < 65, PINCP $\leq 30{,}000$: **138,554** people, **85,182** households; $P(Y{=}1) = 0.631$.
- Protected: recorded race (9 codes), sex, disability $\Rightarrow$ 36 intersections.
- Split 60/20/20 **by household** (SERIALNO), 5 seeds. Features: the 19 Folktables inputs.
- Small groups: only **21 of 35** test intersections have ≥ 30 positives; the rest are exploratory.

Same BN decisions, seed 0 test:

Grouping	groups ($\geq$30)	ΔEOpp
Sex	2 (2)	0.162
Disability	2 (2)	0.384
Race	9 (8)	0.597
Sex$\times$Dis	4 (4)	0.468
Race$\times$Sex$\times$Dis	35 (21)	**0.746**

Intersections expose a gap that one attribute hides (Experiment 1).

Graphical model and inference



$$p(y, \mathbf{x}) = p(y) \prod_j p(x_j \mid \mathrm{pa}(x_j))$$

- Augmented naive Bayes: $Y \to$ all 18 features, plus 10 feature edges (e.g. DIS|AGEP); Dirichlet $\alpha = 1$ CPTs. **Associational, not causal.**
- Query $P(Y{=}1 \mid \mathbf{x}_E)$ by exact variable elimination; held-out codes clipped to the training range.
- Validation: our VE (2 orders), pgmpy VE, clique tree and enumeration agree to $\mathbf{3.3 \times 10^{-16}}$ over 200 evidence masks (tolerance 10^{-9}); joint CI $\mathrm{SEX} \perp \{\mathrm{AGEP}, \mathrm{DIS}\} \mid Y$ checked.
- Test AUROC 0.729 (LR 0.716, GBT 0.768).

Fairness criterion and proposed method

- **Primary criterion:** intersectional equal opportunity (TPR range). The harm is missing people who need outreach.
- **Utility:** recall at 30% capacity; allowed loss ≤ 0.02 (preregistered).
- **Audit model:** $k_g \sim \text{Bin}(n_g, \theta_g)$ for selected positives. Comparators: raw, independent Beta, complete, exchangeable EB / full Bayes, `struct` (logit main effects $+ u_g$, $u_g \sim N(0, \tau^2)$).
- **Proposed:** contamination prior $\theta_g \sim \pi\,\text{Beta}(a, b) + (1 - \pi)\,\text{slab}$, with $\pi_{\max}$, slab and rule chosen by $\max J = \text{ACC} - \overline{\text{FTS}}$ on worlds built on the **target state's validation group sizes**. Then choose λ on validation half A and **certify on half B**.
- *A* **at prediction time:** yes. The scores use race, sex and disability; for $\lambda > 0$ so do the thresholds.
- **Diagnostics:** ΔDP, ΔEOdds, selection ratio, worst-group recall, group ECE, PPC, interval coverage.

Experimental protocol

- **Baselines:** LR, GBT (predictive); BN without uncertainty-aware selection (graphical); raw, independent, complete, hier EB, full Bayes, `struct` audits (fairness).
- **Matched:** same counts for every audit; same capacity (matched on validation); same rule family $R(\lambda)$, $\lambda \in \{0, 0.1, \ldots, 1\}$.
- **Selection:** train fits edges and CPTs; validation picks thresholds and λ; test only evaluates frozen rules. Calibration of the audit uses simulated outcomes only.
- **Runs and intervals:** 5 household splits; household-cluster Poisson bootstrap $B = 1000$, Holm over 2 confirmatory comparisons; paired ΔJ bootstrap over simulated worlds ($B = 2000$).
- **Mechanism tests:** E1–E5, minority thinning (1, 0.5, 0.25, 0.1), semi-synthetic deficit worlds (cell, interaction, disability main effect), prior/tolerance grids, survey weights, Texas replication.

Main result and uncertainty

Fresh Texas worlds (200 per cell), $\epsilon = 0.10$

Audit	ACC↑	FTS↓	J ↑
Full Bayes, plug-in	1.000	0.615	0.385
Hier EB, upper 0.9	0.985	0.625	0.360
`struct`, plug-in	0.108	0.073	0.034
Design-matched	0.832	0.168	**0.664**

Real CA+TX (10 splits)	claims	false
Same-data certification	10	**10**
Held-out certification	0	**0**

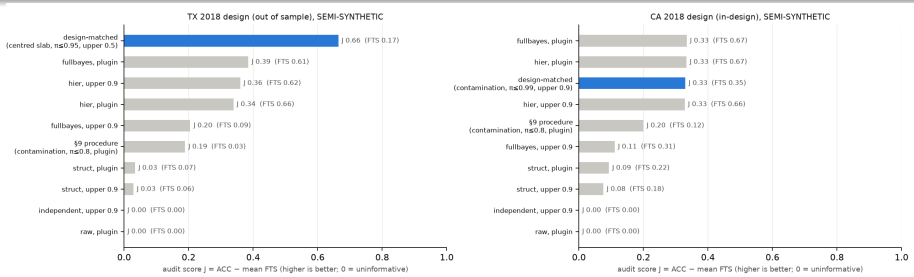
Mean test recall 0.365 in both.

Paired ΔJ (95% CI):
vs best established: $+0.279$ **[0.235, 0.322]**
vs earlier calibration: $+0.475$ [0.422, 0.525]

H1: 116/116 CA and **27/27** TX pooled validation claims failed on test $\Rightarrow$ supported.

Answer: pooled "gap ≤ 0.10" claims are not trustworthy for sparse intersections. Calibrating on the real design and certifying on held-out households gives the best audit out of sample on Texas; on California it ties.

Ablation, robustness and failure case



Audit score $J = \text{ACC} - \text{mean FTS}$ on the Texas (left, out of sample) and California (right, pilot) designs.

- **Ablation, no design matching:** fair-world acceptance 0.83 on invented sizes $\rightarrow$ 0.32 (CA) and 0.22 (TX); 90% of false alarms come from groups with < 10 positives. **Same-data certification:** 10/10 claims, all false.
- **Robustness:** EB upper 0.9 certifies 98% of worlds with one hidden 20-positive deficit; a disability main-effect deficit is caught by every audit. **Uncertainty:** bootstrap = evaluation, 5 refits = training (mixed signs), simulation = scenario.
- **Failure case:** on the CA design J ties (-0.004 [-0.057, 0.046]); no rule is fair on test (min gap 0.227); the preregistered real-gap comparison is inconclusive (Holm $p = 0.736$).

Conclusion and reproducibility

- **Conclusion:** pooling masks isolated small-group deficits, and pooled validation-time claims failed on test in every case (H1). Design-matched calibration with held-out certification gives the best audit score out of sample on Texas ($J = 0.664$ vs at most 0.385) and withheld every false real-data claim.
- **Limits and ethics:** simulation evidence comes from one generator family; zero held-out claims shows abstention, not sensitivity; the label is not need; no stakeholder set ϵ. Do not call any rule here fair.
- **Reproduce:** `https://github.com/sumedhnvda/ict5143-p8-fairness-audits`, tag `submission-v1`; `jupyter nbconvert --execute --inplace Problem8_Intersectional_Fairness.ipynb` (about 30 min, 1 CPU).
- **Sumedh Navuda:** question, literature, data, BN and exact inference, Texas H1 test; cross-checked the audits and J.
- **Aldino:** audit models, simulations, threshold selection, bootstrap, design-matched/held-out audit (H2); cross-checked splits, labels and inference.

Backup: mandatory evidence and references

Requirement	Notebook	Status / artefact
VE vs clique tree, normalisation, CI	§3.1	complete; max error 3.3×10^{-16}
E1 single vs intersectional	§5	complete; `exp1_single_two_three_way.csv`
E2 minority thinning	§10	complete; 4 levels $\times$ 5 seeds, `selection_all.csv`
E3 empirical vs independent vs pooled	§7	complete; `exp3_estimates_real.csv`
E4 homogeneous vs exceptional worlds	§8, §9.2	complete; `sim_main_summary`, `semisynthetic_*`
E5 ordinary vs uncertainty-aware selection	§10	complete; capacity matched (test 0.292–0.305)
Ablations: no / complete pooling, mean vs quantile	§8, §10	complete
Robustness: priors, prevalence, ϵ, 2nd state	§9.1, §12, §14	complete; TX replication
Bootstrap $\geq$1000, Holm, $\geq$5 seeds	§11	complete; `confirmatory_bootstrap.csv`
H1 / H2a–c	§14.1, §15	supported; `h1_texas_test`, `v2_*`

Full matrix: `docs/requirements_matrix.md`. References: Foulds et al. SDM 2020; Herlihy et al. FAccT 2024; Kearns et al. ICML 2018; Berger & Berliner, Ann. Stat. 1986; Ding et al. NeurIPS 2021; Cotter et al. ICML 2019 (19 in `writeup/references.tex`).

Backup: inference and statistical validation

- **Factors:** scope and shape validated on construction; evidence reduces by slicing, hidden variables are summed; final table normalised. CPT rows sum to 1 ($\alpha = 1$ keeps every entry positive).
- **Exact inference:** 6-node sub-network, 200 masked-evidence cases: our VE (2 orders), pgmpy VE, clique tree, enumeration agree to 3.3×10^{-16}. The full-BN fast scorer equals general VE on observed rows.
- **Independence:** for each y, $P(\text{SEX}, \text{AGEP}, \text{DIS} \mid y) = P(\text{SEX} \mid y) P(\text{AGEP}, \text{DIS} \mid y)$.
- **`struct` audit:** quadrature vs fine grid 9.6×10^{-11}; analytic vs numeric gradient 2.0×10^{-6}; 90% coverage 0.665 fitted vs 0.905 oracle ($\hat{\tau}$ 0.18 vs true 0.30).
- **Denominators:** positives per group 1–5,499 on validation; groups with < 30 positives flagged exploratory; zero-positive groups have undefined TPR.
- **Bootstrap:** Poisson(1) weight per household, shared by both arms; percentile CI; Holm over {proposed vs hier plug-in, proposed vs raw plug-in}: $p_{\text{Holm}} = 1.0$, 0.736.

Backup: demonstration and individual defence

- **Live demo (about 15 s):** `python writeup/make_tables.py` regenerates every table and the paired ΔJ intervals from `results/`.
  Expected line: `TX ... fullbayes · plugin ... 0.279 0.235 0.322`.
- **Full regeneration:** execute the notebook (about 30 min, 1 CPU core, downloads 435 MB of ACS data).
- **Saved fallback trace:** `demo/saved_trace.txt` (inference check, H1 verdict, H2 scores).
- **Trace an inference:** a factor with axes (B, A), shape $(3, 2)$, times (B, C), shape $(3, 4)$, broadcasts to $(2, 3, 4)$; summing B removes its axis; normalising gives $P(\cdot \mid e)$.
- **Justify fairness:** TPR range, because the harm is unreached people who need outreach; $\epsilon = 0.10$ is a value judgement tested at 0.05 and 0.15.
- **Critique a claim:** zero held-out false claims follows from zero claims; it shows abstention, not that the procedure would certify a really fair rule.